

Part I

SUMMARY of Product Rule and Quotient Rule

Summary

February 21, 2026

Summary

- **The Product Rule:** If f and g are differentiable functions, then $\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$.
- **The Quotient Rule:** If f and g are differentiable functions, then $\frac{d}{dx} \frac{f(x)}{g(x)} = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$.

Part II

Recitation Questions

Problem 1

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Problem 1 Differentiate the functions using product or quotient rule.

(a) $h(u) = 7ue^u$

Explanation. Applying the power rule:

$$\begin{aligned} h(u) &= 7ue^u \\ \implies h'(u) &= (7ue^u)' \\ &= 7 \cdot e^u + 7u \cdot e^u \end{aligned}$$

(b) $s(t) = \frac{\sqrt{t}}{e^t}$

Explanation. Applying the quotient rule:

$$\begin{aligned} s(t) &= \frac{\sqrt{t}}{e^t} \\ \implies s'(t) &= \left(\frac{\sqrt{t}}{e^t} \right)' \\ &= \frac{\frac{1}{2}t^{-1/2} \cdot e^t - \sqrt{t} \cdot e^t}{e^{2t}} \end{aligned}$$

Problem 2

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Problem 2

Problem 2 Differentiate the function f defined by $f(x) = 1/x^8$ in two different ways.

Explanation. Applying the quotient rule:

$$\begin{aligned} f'(x) &= \frac{(1)' \cdot x^8 - (1 \cdot (x^8)')}{(x^8)^2} \\ &= \frac{(0 \cdot x^8) - (1 \cdot 8x^7)}{x^{16}} \\ &= \frac{-8x^7}{x^{16}} \\ &= \frac{-8}{x^9}. \end{aligned}$$

Applying the power rule:

$$\begin{aligned} f(x) &= \frac{1}{x^8} = x^{-8} \\ \implies f'(x) &= (x^{-8})' \\ &= -8 \cdot x^{-8-1} \\ &= -8x^{-9} \end{aligned}$$

Problem 3

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Problem 3 Suppose that $f(5) = 7$, $f'(5) = 8$, $g(5) = 3$, and $g'(5) = -4$. Find:

(a) $(fg)'(5)$.

Explanation.

$$\begin{aligned}(fg)'(5) &= (f'(5) \cdot g(5)) + (f(5) \cdot g'(5)) \\ &= (8)(3) + (7)(-4) \\ &= 24 - 28 = -4.\end{aligned}$$

(b) $\left[\frac{d}{dx} \left(\frac{f}{g} \right) \right]_{x=5}$

Explanation.

$$\begin{aligned}\left[\frac{d}{dx} \left(\frac{f}{g} \right) \right] (5) &= \frac{(g(5) \cdot \left[\frac{d}{dx} f \right] (5)) - (f(5) \cdot \left[\frac{d}{dx} g \right] (5))}{(g(5))^2} \\ &= \frac{(3)(8) - (7)(-4)}{3^2} \\ &= \frac{24 + 28}{9} = \frac{52}{9}.\end{aligned}$$

(c) $\left(\frac{g}{f} \right)' (5)$

Explanation.

$$\begin{aligned}\left(\frac{g}{f} \right)' (5) &= \frac{(f(5) \cdot g'(5)) - (g(5) \cdot f'(5))}{(f(5))^2} \\ &= \frac{(7)(-4) - (3)(8)}{7^2} \\ &= \frac{-28 - 24}{49} = -\frac{52}{49}.\end{aligned}$$

(d) $\left[\frac{d}{dx} \left(\frac{g(x)}{x+2} \right) \right]_{x=5}$

Explanation.

$$\begin{aligned}\left[\frac{d}{dx} \left(\frac{g(x)}{x+2} \right) \right]_{x=5} &= \left[\frac{(x+2) \cdot g'(x) - (g(x) \cdot (x+2)')}{(x+2)^2} \right]_{x=5} \\ &= \frac{(5+2)(-4) - (3)(1)}{(5+2)^2} \\ &= \frac{(7)(-4) - (3)(1)}{7^2} \\ &= \frac{-28 - 3}{49} = -\frac{31}{49}\end{aligned}$$

$$(e) \left[\frac{d}{dx} \left(\frac{xf(x)}{g(x)} \right) \right]_{x=5}$$

Explanation.

$$\begin{aligned} \left[\frac{d}{dx} \left(\frac{xf(x)}{g(x)} \right) \right]_{x=5} &= \left[\frac{(((xf(x))' \cdot g(x)) - (g'(x) \cdot (xf(x))))}{(g(x))^2} \right]_{x=5} \\ &= \left[\frac{((x)' \cdot f(x) + x \cdot f'(x)) \cdot g(x) - (g'(x) \cdot (xf(x)))}{(g(x))^2} \right]_{x=5} \\ &= \frac{(1 \cdot 7 + 5 \cdot 8)(3) - (-4)(5)(7)}{(3)^2} \\ &= \frac{141 + 140}{9} \\ &= \frac{281}{9} \end{aligned}$$

$$(f) \lim_{x \rightarrow 5} \frac{x^2 g(x) - 75}{x - 5}. \text{ **EXPLAIN.**}$$

Explanation. We first note that this limit has form $\frac{0}{0}$, but with no formula for $g(x)$ there is no simplification we can perform. Notice that if we set $w(x) = x^2 g(x)$, then $w(5) = (5)^2 g(5) = 75$. This limit is the definition of the derivative, $w'(5)$.

$$\begin{aligned} \lim_{x \rightarrow 5} \frac{x^2 g(x) - 75}{x - 5} &= w'(5) \\ &= \left[\frac{d}{dx} (x^2 g(x)) \right]_{x=5} \\ &= \left[2xg(x) + x^2 g'(x) \right]_{x=5} \\ &= 2(5)g(5) + (5)^2 g'(5) \\ &= -70 \end{aligned}$$

$\lim_{x \rightarrow 5} \frac{x^2 g(x) - 75}{x - 5} = -70$ by the product rule since this is the definition of the derivative of $x^2 g(x)$ at $x = 5$.

$$(g) \lim_{x \rightarrow 5} \frac{\frac{x+2}{f(x)} - 1}{x - 5}. \text{ **EXPLAIN.**}$$

Explanation. We first note that this limit has form $\frac{0}{0}$. Notice that if we set $k(x) = \frac{x+2}{f(x)}$, then $k(5) = \frac{5+2}{f(5)} = 1$. This limit is the definition of

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the derivative, $k'(5)$.

$$\begin{aligned}\lim_{x \rightarrow 5} \frac{\frac{x+2}{f(x)} - 1}{x - 5} &= k'(5) \\ &= \left[\frac{d}{dx} \left(\frac{x+2}{f(x)} \right) \right]_{x=5} \\ &= \left[\frac{f(x) - (x+2)f'(x)}{(f(x))^2} \right]_{x=5} \\ &= \frac{f(5) - (5+2)f'(5)}{(f(5))^2} \\ &= \frac{7 - 7(8)}{(7)^2} \\ &= -1\end{aligned}$$

$\lim_{x \rightarrow 5} \frac{\frac{x+2}{f(x)} - 1}{x - 5} = -1$ by the quotient rule since this is the definition of the derivative of $\frac{x+2}{f(x)}$ at $x = 5$.

Problem 4

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Problem 4 Use the given information to find the equation of the tangent line.

- (a) Given $g(x) = x^3 f(x)$, $f(2) = 4$, and $f'(2) = 7$, find the equation of the tangent line to the graph of g at $x = 2$.

Explanation. Slope of tangent line to the graph of g at the point where $x = 2$:

$$\begin{aligned} g'(x) &= 3x^2 f(x) + x^3 f'(x) \\ \implies g'(2) &= 12(4) + 8(7) = 48 + 56 = 104. \end{aligned}$$

Point on tangent line where $x = 2$: $(2, g(2)) = (2, 8f(2)) = (2, 32)$.

Equation of tangent line:

$$\begin{aligned} y - 32 &= 104(x - 2) \\ \implies y &= 104x - 176. \end{aligned}$$

- (b) Given $h(z) = \frac{zs(z)}{z-3}$, $s(2) = 4$, and $s'(2) = 7$, find the equation of the tangent line to the graph of h at $z = 2$.

Explanation. Slope of tangent line to the graph of h at the point where $z = 2$:

$$\begin{aligned} h'(z) &= \frac{(z-3)(zs(z))' - zs(z)(z-3)'}{(z-3)^2} \\ &= \frac{(z-3)(s(z) + zs'(z)) - zs(z)(1)}{(z-3)^2} \\ \implies h'(2) &= \frac{(2-3)(s(2) + 2s'(2)) - 2s(2)}{(2-3)^2} = -26. \end{aligned}$$

Point on tangent line where $z = 2$: $(2, h(2)) = (2, \frac{2s(2)}{2-3}) = (2, -8)$.

Equation of tangent line to the graph of h at the point where $z = 2$:

$$\begin{aligned} y - (-8) &= -26(z - 2) \\ \implies y &= -26z + 44. \end{aligned}$$

- (c) Given

x	1	2	3	4	5
$f(x)$	5	3	0	-4	3
$f'(x)$	-3	-5	-2	6	-4
$g(x)$	6	9	-8	13	15
$g'(x)$	8	5	-10	7	6

find the equation of the tangent line of

$$\frac{f(x)}{e^x g(x)}$$

at $x = 2$.

Explanation. Slope of tangent line to the graph of $\frac{f(x)}{e^x g(x)}$ at the point where $x = 2$:

$$\begin{aligned} \left(\frac{f(x)}{e^x g(x)} \right)' &= \frac{e^x g(x) f'(x) - f(x)(e^x g(x) + e^x g'(x))}{(e^x g(x))^2} \\ &\Rightarrow \frac{d}{dx} \left(\frac{f(x)}{e^x g(x)} \right) \Big|_{x=2} = \frac{e^2 g(2) f'(2) - f(2)(e^2 g(2) + e^2 g'(2))}{(e^2 g(2))^2} \\ &= \frac{e^2(9)(-5) - (3)(9e^2 + 5e^2)}{(9e^2)^2} \\ &= \frac{-45e^2 - 42e^2}{81e^4} \\ &= \frac{-87e^2}{81e^4} \\ &= \frac{-87}{81e^2} \end{aligned}$$

Point on tangent line where $x = 2$: $(2, f(2)/(e^2 g(2))) = (2, 3/(9e^2)) = (2, 1/(3e^2))$.

Equation of tangent line where $x = 2$:

$$\begin{aligned} y - \frac{1}{3e^2} &= \frac{-87}{81e^2}(x - 2) \\ \Rightarrow y &= \frac{-87}{81e^2}x + \frac{67}{27e^2}. \end{aligned}$$

Problem 5

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Problem 5 Differentiate the following functions:

- (a) Given $f(x) = (x^2 + 4x - 7)e^{-x}$, show $f'(x) = (11 - x^2 - 2x)e^{-x}$.

Explanation.

$$\begin{aligned} f'(x) &= \frac{d}{dx} \left(\frac{x^2 + 4x - 7}{e^x} \right) \\ &= \frac{(2x + 4)e^x - (x^2 + 4x - 7)e^x}{e^{2x}} \\ &= \frac{e^x(2x + 4 - x^2 - 4x + 7)}{e^{2x}} \\ &= \frac{11 - x^2 - 2x}{e^x} \\ &= (11 - x^2 - 2x)e^{-x} \end{aligned}$$

- (b) Given $g(x) = \frac{x^2 + 4x - 7}{e^{-x}}$, show $g'(x) = \frac{x^2 + 6x - 3}{e^{-x}}$

Explanation.

$$\begin{aligned} g'(x) &= \frac{d}{dx} \left(\frac{x^2 + 4x - 7}{e^{-x}} \right) \\ &= \frac{d}{dx} \left((x^2 + 4x - 7)e^x \right) \\ &= \frac{d}{dx} (x^2 + 4x - 7)e^x + (x^2 + 4x - 7) \frac{d}{dx} e^x \\ &= (2x + 4)e^x + (x^2 + 4x - 7)e^x \\ &= e^x(2x + 4 + x^2 + 4x - 7) \\ &= e^x(x^2 + 6x - 3) \\ &= \frac{x^2 + 6x - 3}{e^{-x}} \end{aligned}$$

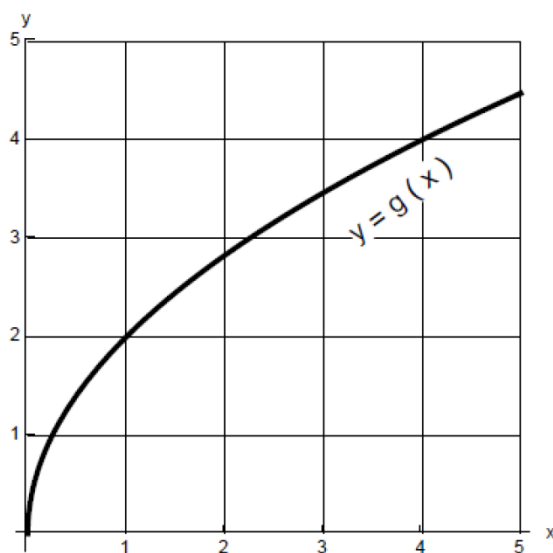
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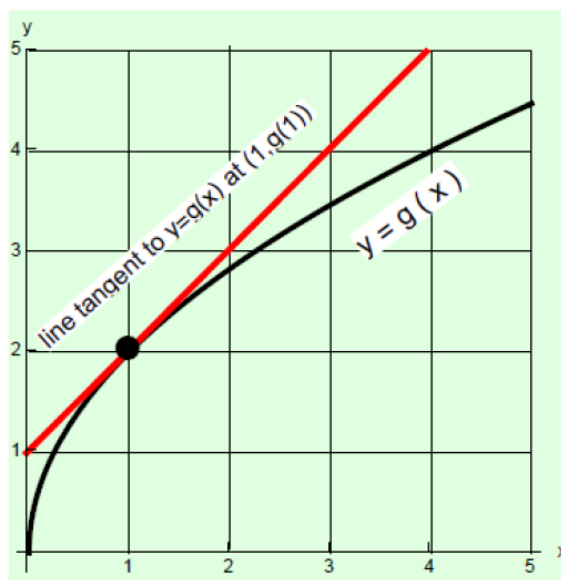
Problem 6

Problem 6 The graph of a function g is given below. Using the graph, estimate the derivative at the given point:

$$\frac{d}{dx} \left(\frac{xg(x)}{x+3} \right) \text{ at } x = 1$$



Explanation. From the graph it appears that $g(1) = 2$. We can draw the line tangent to the curve $y = g(x)$ at the point $(1, g(1))$



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Since $g'(1)$ is the slope of the line tangent to the curve $y = g(x)$ at the point $(1, g(1))$, we can estimate it from the figure above. The tangent line seems to pass through the points $(1, 2)$ and $(2, 3)$, so its slope is $\frac{3-2}{2-1} = 1$.

$$\begin{aligned}
 \left[\frac{d}{dx} \left(\frac{xg(x)}{x+3} \right) \right]_{x=1} &= \left[\frac{(g(x) + xg'(x))(x+3) - xg(x)}{(x+3)^2} \right]_{x=1} \\
 &= \frac{(g(1) + 1g'(1))(1+3) - (1)g(1)}{(1+3)^2} \\
 &\approx \frac{(2 + 1(1))(4) - 1(2)}{4^2} \\
 &= \frac{(2+1)(4) - 2}{16} \\
 &= \frac{3(4) - 2}{16} \\
 &= \frac{12 - 2}{16} \\
 &= \frac{10}{16} \\
 &= \frac{5}{8}
 \end{aligned}$$